ps1

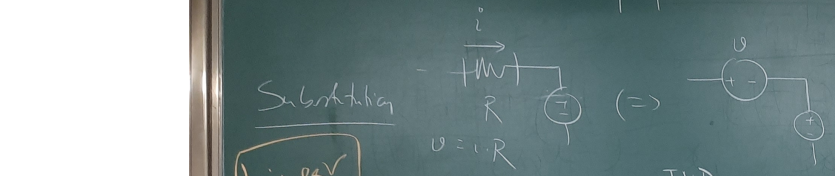
Recap

Substitution thm

Superposition thm

} Come from the linear eqns (mesh / nodal)

Substitution



Linear

Superposition: Activate one source at a time (deactivate all other sources)

VS → short

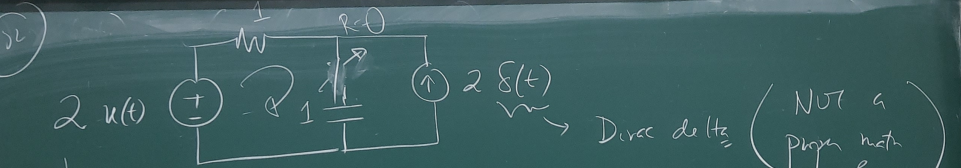
CS → open

IND

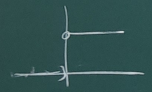
NW

⇒ get curr/volt in each case ⇒ sum

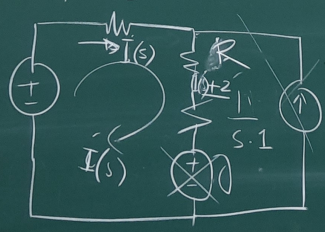
ps2



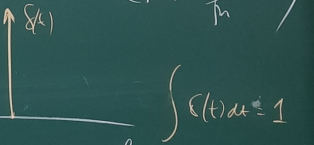
Dirac delta (NOT a proper math fn)



Laplace



$\frac{2}{s}$



$\int \delta(t)\,dt = 1$

$\delta(t) \xrightarrow{\mathcal{L}} 1$

$\delta(t-t_0)$

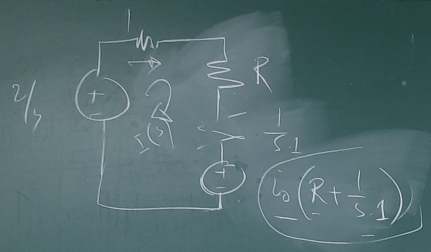
PS2

$$-\frac{2}{s} + I(s)\cdot 1 + \left(I(s)+i_0\right)\cdot R + \left(I(s)+i_0\right)\frac{1}{s\cdot 1} = 0$$

$$-\frac{2}{s} + I(s)\cdot 1 + I(s)\cdot R + I(s)\cdot\frac{1}{s\cdot 1} + i_0\left(R+\frac{1}{s\cdot 1}\right) = 0$$

$$\frac{1}{s}\, i_0\left(sR+1\right)$$

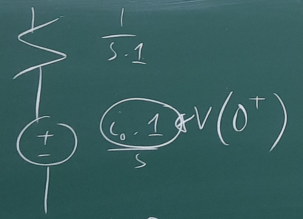
2/s



2PS4

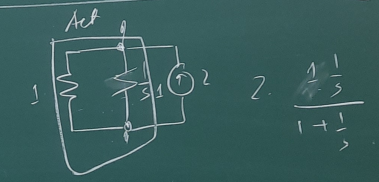
$R=0$

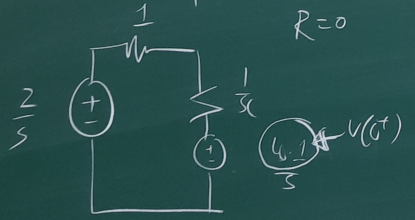
$$\frac{i_0\cdot 1}{s}$$



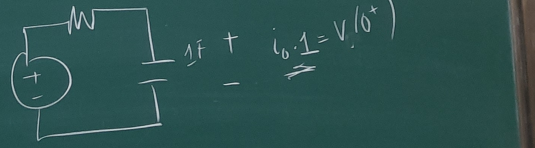
Act Vs

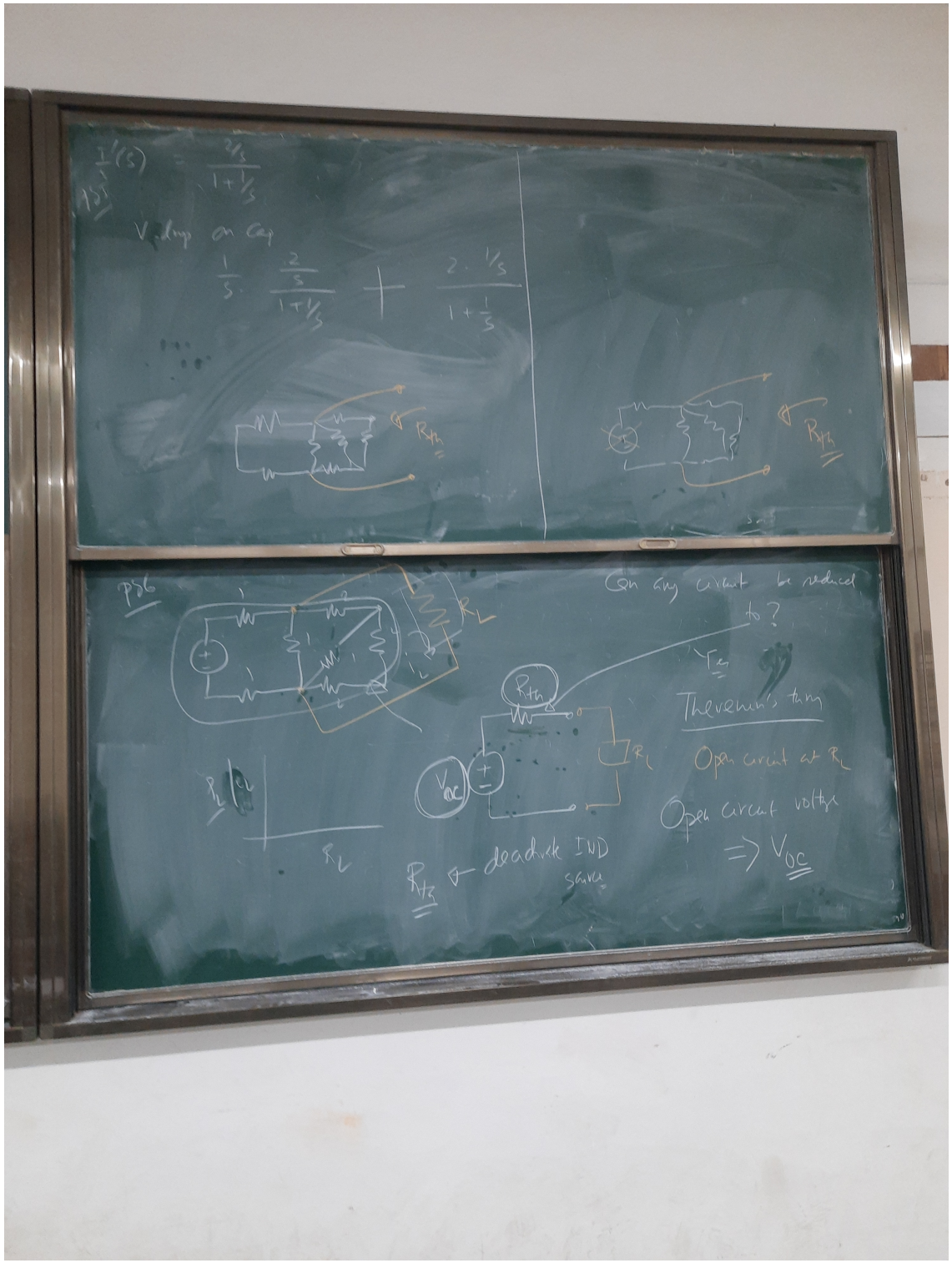
$$-\frac{2}{s} + I'(s)\cdot 1 + I'(s)\cdot\frac{1}{s\cdot 1} = 0$$

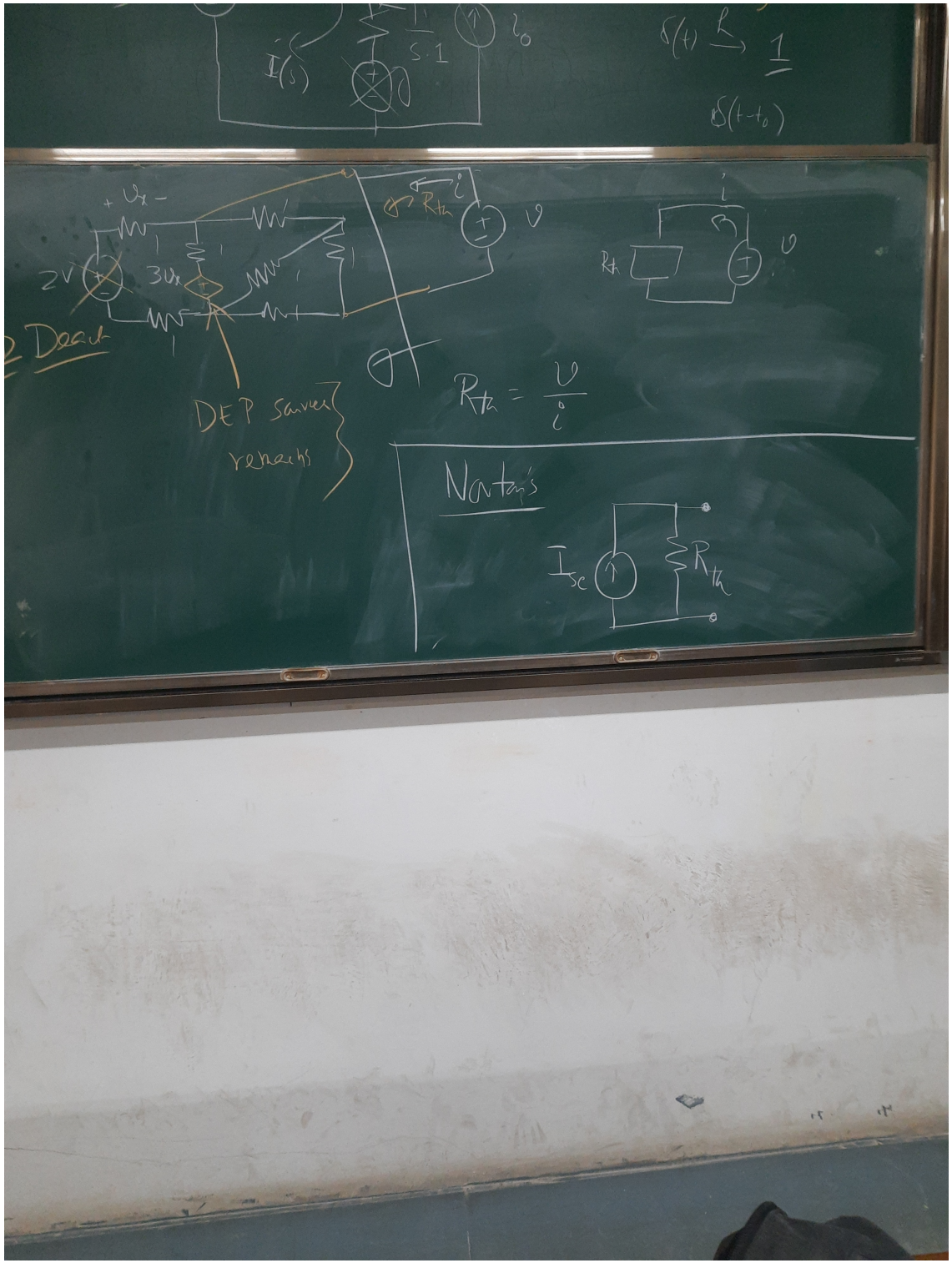




$R=0$




Voltage on cap
R_{Th}
R_{Th}
P56
R_L
Can any circuit be reduced to ?
Thevenin's thm
Open circuit at R_L
Open circuit voltage
$\Rightarrow V_{OC}$
R_{Th}
R_L
V_{OC}
R_L
R_{Th} ← deactivate IND sources


$I(s)$
$s\cdot 1$
i_o
$\delta(t) \xrightarrow{\mathcal{L}} 1$
$\delta(t-t_0)$
$+\ v_x\ -$
$2V$
$3v_x$
R_{th}
i
v
R_{th}
i
v
2 Deact
DEP sources remain
$R_{th} = \frac{v}{i}$
Norton's
I_{sc}
R_{th}